

DermaLens AI

Clinical question: should this skin lesion be flagged for closer clinical assessment?

Proposed allocation: about 103 GPU-hours on Leonardo Booster; 4 GPUs core, 8-GPU benchmark conditional

Politecnico di Milano | Supervisor: Alessio Ronchini | Code: github.com/pr4jtk8fhj-dotcom/derm-pretraining-adaptation

1. Goal, clinical partner and measurable scope

We are developing an image-based research system that flags dermoscopic skin lesions for closer assessment by a clinician. A positive flag prompts review; it does not prescribe biopsy. A negative flag is not a recommendation to omit clinical assessment or follow-up.

Clinical partner. The project is carried out under a non-disclosure agreement with an Italian hospital specialised in dermatology. Ethics approval for the hospital study was granted in October 2026. Hospital images will never be transferred to Leonardo: after the hackathon, the locked model, preprocessing and threshold will be installed and evaluated inside the hospital.

Central question. Does continued self-supervised pretraining on public dermoscopy improve external diagnostic discrimination relative to original DINOv2, and at what computational cost? Leonardo is the computational foundry in which the encoder is trained, profiled and evaluated; the resulting checkpoint, not the supercomputer, is the asset intended for later local use.

2. Models and pre-specified research questions

- **DINOv2 ViT-L/14.** General-purpose vision encoder of about 300 million parameters; Apache-2.0 code and weights. It is the base encoder.

- **DINOv2-CPT.** The same architecture after continued pretraining on unlabelled public dermoscopic images using the official DINOv2 trainer and DINO, iBOT and KoLeo losses. This is the encoder to be produced on Leonardo.

- **PanDerm ViT-L/16.** Dermatology foundation model used as an external reference. Its architecture, training history and licence differ, so differences cannot be attributed only to domain pretraining.

Primary contrast. DINOv2-CPT minus original DINOv2 under full fine-tuning at 224 px with 100% of annotations; endpoint: paired difference in external AUROC.

Secondary questions. How does performance vary across frozen features, LoRA and full fine-tuning; across 10%, 30% and 100% of annotations; and between 224 px and 448 px? Which execution strategy reduces elapsed time and resource use while preserving a scientifically comparable experiment?

3. Public data and independence of evaluation

All development images come from the ISIC Archive. Public data contain no flagging labels; we therefore measure a diagnostic proxy: histopathology-confirmed malignant versus benign. The labelled cohort contains 39,332 dermoscopic images. Indeterminate and missing diagnoses were excluded rather than relabelled.

Subset	Images	Malignant / benign	Patients	Sources
Labelled training, 10%	3,023	1,616 / 1,407	228	11
Labelled training, 30%	9,072	4,851 / 4,221	695	11
Labelled training, 100%	30,250	16,184 / 14,066	2,358	12
Validation	5,042	2,698 / 2,344	394	11
External test: HIBA + MSKCC	2,784	1,014 / 1,770	1,455	2
Continued-pretraining pool	45,742	unlabelled	4,153	10

External evaluation and leakage control. HIBA and the 2,045 patient-identified MSKCC images form the external test used in Phase A and retained for the controlled Phase B comparison. Exact and perceptual duplicate audits link 1,267 near-duplicate images; patient, lesion and duplicate groups never cross partitions. The continued-pretraining pool contains 18,103 CC-0 and 27,639 CC-BY images. A remaining limitation is that 40% of pool images have anonymous attribution, so every unverifiable overlap is reported explicitly.

Declared deviations. MSKCC has 63.5% patient-identifier coverage, below our 90% rule, so only its 2,045 identified images are used and the remaining 1,174 are excluded from all sets; HIBA is a benchmark in the PanDerm paper; MSKCC patients may appear in ISIC 2024, part of PanDerm’s pretraining data.

4. Controlled comparison and evaluation protocol

Strategy	Parameters updated	Scientific role / main cost
Frozen features + logistic regression	Classifier only; features extracted once	Measures information already present in the representation
LoRA, rank 16	Low-rank adapters on attention qkv/output projections and classifier (2,361,346 trainable; 2.36 M)	Efficient adaptation; backbone activations still dominate memory
Full fine-tuning	All backbone parameters and classifier	Maximum adaptation capacity; highest gradient and optimizer memory

All arms use the same eligible images, partitions, labels, evaluation code and preprocessing. Only learning rate is tuned, on validation only, with the same number of trials per arm. Thresholds are selected on validation to target 95% sensitivity for malignancy and then locked. External AUROC, achieved sensitivity, specificity, false negatives and flagged fraction are reported by institution and pooled, with 1,000 patient-clustered bootstrap resamples where identifiers permit.

Test discipline. External results do not select hyperparameters, checkpoints, thresholds or priorities. The hackathon reuses the same external cohort, so the subsequent hospital cohort is the next independent evaluation.

5. Phase A evidence: completed on one NVIDIA H100 80 GB

Phase A completed 46 test-evaluated runs, none failed: 6 frozen-feature runs; 36 LoRA and full fine-tuning runs at 224 px (2 encoders x 2 methods x 3 annotation fractions x 3 seeds); and 4 full fine-tuning runs at 448 px. A further 16 learning-rate trials used validation data only.

Encoder	Adaptation	10%	30%	100%
PanDerm	Frozen	0.791	0.803	0.831
PanDerm	LoRA r16	0.855	0.863	0.884
PanDerm	Full	0.840	0.846	0.862
DINOv2	Frozen	0.697	0.763	0.802
DINOv2	LoRA r16	0.855	0.862	0.885
DINOv2	Full	0.848	0.859	0.879

Pooled external AUROC at 224 px; means across seeds for LoRA and full fine-tuning. Full sensitivity, specificity, confidence intervals and per-institution results are retained in the repository.

- **Adaptation benefit.** LoRA and full fine-tuning exceeded frozen features in all 36 paired comparisons; every interval excluded zero.
- **Efficiency signal.** LoRA matched or exceeded full fine-tuning (full minus LoRA between -0.033 and 0.000), but the selected full fine-tuning learning rate lay at the edge of the grid for both encoders, so this comparison may favour LoRA.
- **Resolution signal.** DINOv2 full fine-tuning at 448 px improved over 224 px for all three seeds, reaching AUROC 0.892 (SD 0.001). PanDerm at 448 px (one seed) decreased by 0.013 [-0.027, -0.000] to 0.848. No learning-rate sweep was run at 448 px.
- **Open question.** The primary DINOv2-CPT versus DINOv2 comparison remains a Phase B deliverable. Phase A does not claim that continued pretraining already improves downstream performance.

Main limitations. Threshold transfer was imperfect between institutions; MSKCC was consistently harder than HIBA; 448 px runs were not compute-matched; and the PanDerm comparison is subject to the declared benchmark and pretraining-overlap caveats.

6. Computational profile and measured bottlenecks

Measurement	Result	Interpretation
DINOv2-CPT smoke test	170.3 original images/s; 47.2 GiB peak; loss 14.03 to 12.88; 2,000 iterations in 10 min	Stable engineering check, not a trained domain model
224 px adaptation	DINOv2 LoRA 639 img/s, 23.7 GB; full 498 img/s, 37.5 GB	LoRA reduces trainable state, but activation cost remains material
Input pipeline (PanDerm full fine-tuning at 224 px)	Cache 77.1 s/epoch and 10.6% idle; original JPEGs 124.0 s and 44.7% idle; DALI 176.3 s and 65.8% idle	Current DALI path is slower; decoding and staging require redesign
448 px full fine-tuning	Time/image x5.8 for DINOv2 and x10.4 for PanDerm; activation checkpointing required	Higher resolution is feasible but must be prioritised after the core contrast

Correctness. LoRA zero-initialisation reproduced both original encoders exactly; gradients reached only the adapters; PanDerm weight loading and weight-decay groups matched the original script; and the continued-pretraining teacher matched the DINOv2 weights within the documented tolerance. Multi-GPU scaling and A100 measurements remain to be completed.

7. Why Leonardo and NVIDIA mentor support are needed

The project does not claim that one baseline is impossible on a single high-end GPU. The hackathon is needed to transform a working single-GPU prototype into a reproducible distributed workflow, quantify scaling and communication overhead, remove measured input bottlenecks, and complete the domain-adaptation experiment under a controlled protocol.

Mentor support is requested for Nsight profiling, FSDP and NCCL behaviour on Leonardo, DALI multi-crop design for DINOv2, checkpoint restart and Slurm-compatible execution. A four-GPU node is the core target; an eight-GPU, two-node benchmark is conditional on measured single-node benefit.

8. Hackathon work plan and decision gates

Work package	Activity and decision
B1 - Port and profile	Reproduce one complete cycle on A100; profile input, attention and communication; validate checkpoint restart under Slurm.
B2 - Scale continued pretraining	Benchmark 1 and 4 GPUs; add an 8-GPU benchmark only if single-node results justify it. Measure strong and weak scaling.
B3 - Optimise input	Compare CPU decoding, reusable preprocessing and DALI/nvJPEG while preserving field of view and augmentation semantics. Target less than 10% training idle time.
B4 - Complete and evaluate DINOv2-CPT	Fix the training schedule from validation and B1 evidence; select checkpoints on validation only; run the primary contrast at 224 px, then prioritise 448 px and annotation-scaling analyses.

9. Resource request

For continued pretraining, GPU-hours = $g \times M / (3,600 \times Qg)$, where g is allocated GPUs, M scheduled image instances and Qg measured aggregate throughput in original images per second. The schedule length and an A100 factor of two are planning assumptions to be replaced by B1 measurements.

Item	Technical basis	Request
Continued pretraining	9.15 million image instances: 200 passes over 45,742 images; H100 Q1 = 170.3 img/s; x2 A100 planning factor	30 GPU-h
Downstream comparisons	DINOv2-CPT learning-rate trials, 224 px matrix, 448 px core runs, and remaining PanDerm 448 px seeds	20 GPU-h
Profiling and scaling	1, 4 and conditional 8 GPU timing windows; porting, Nsight profiling, restart tests and communication analysis	36 GPU-h
Validation and contingency	Evaluation, checkpoint I/O and 20% allowance for justified reruns	17 GPU-h
Total	Allocated GPUs multiplied by job walltime, summed across work packages	about 103 GPU-h

Resource envelope. One four-GPU A100 node for core work; two nodes only for a conditional benchmark. Peak H100 memory was 47.2 GiB for continued pretraining, 41.4 GB for PanDerm full fine-tuning at 224 px and 40.8 GB reserved at 448 px. Requested storage is 300 GB; planned walltime is up to eight-hour restartable segments.

10. Hospital validation and transfer

After the hackathon, the locked model selected under the rule below is installed at the partner hospital and evaluated on its own patients under the approved protocol. The cohort is being assembled for this validation study and uses dermoscopic images, histopathology-confirmed benign or malignant diagnosis, consecutive eligible lesions over a defined period, and patient and lesion identifiers. The target includes at least 300 malignant lesions.

Pre-specified model-selection rule. The encoder, adaptation method, resolution and threshold are chosen on public validation data only. Among matched DINOv2 and DINOv2-CPT configurations completed with the planned seed count, the method-resolution pair with the highest mean validation AUROC is selected; exact ties favour the lower-compute option. Within that pair, DINOv2-CPT is installed only if it exceeds original DINOv2 on validation; otherwise original DINOv2 is installed. The threshold is locked on validation to target 95% sensitivity.

Because Phase A showed imperfect threshold transfer, a secondary analysis recalibrates on a separate hospital calibration subset and evaluates on the untouched remainder. Results are reported overall and by device, lesion type, body site, age and sex. Only aggregate metrics leave the hospital.

Why transfer is plausible. We expect transfer because the encoder is adapted on 45,742 dermoscopic images from 10 sources, and Phase A models kept their discrimination on two external institutions in two countries that were never used for Phase A training (DINOv2 full fine-tuning, 224 px: HIBA 0.910-0.928, MSKCC 0.806-0.819). The hospital cohort also uses dermoscopes represented in the training data and the same histopathology endpoint. The lower MSKCC results and imperfect threshold transfer (test sensitivity 0.87-0.96 against a 95% validation target) show that performance can drop at a new site; this is why the protocol includes a separate calibration subset and per-device analysis.

11. Team, affiliation and research readiness

Academic affiliation. The five-member team is affiliated with Politecnico di Milano through the Executive MBA and is supervised by Alessio Ronchini.

Member and background	Practical competence and accountable role
Gaia Arienti - Telecommunications Engineer; Product Manager, Enel Grids & Innovation	Python/PyTorch and Linux/Slurm. Technical and HPC Lead: repository ownership, A100 porting, FSDP/NCCL execution, profiling, debugging and mentor interface.
Olimpia Cordeschi - Corporate Sales Manager, Virgin Active	Run planning, decision logging and stakeholder coordination. Project and Experimental Strategy Lead: objectives, priorities, daily coordination and final presentation.
Alessandra Fatone - Engineer and Architect; Technical Evaluator of Business Projects	Configuration traceability and resource planning. Reproducibility Lead: run register, allocation plan, risks and documentation.
Marco Tarchi - Chemist; Senior Medical Science Liaison, Angelini Pharma	Biomedical literature review and clinical endpoint interpretation. Biomedical Evidence and Translation Lead: clinical framing, endpoint review and hospital-validation interface.
Gianluca Cividini - Chemical Engineer; co-founder, Brainy	Protocol checklists and research documentation. Study Quality Lead: data dictionary, consistency review and final audit trail.

Gaia is the principal technical owner. Before the event, the other four members will complete a shared runbook-based onboarding so that each can reproduce jobs, monitor logs and document results. All five members will attend the full event.

Governance and licences. This is an academic, non-commercial project conducted through Politecnico di Milano; members' employers have no role in it. Project code is Apache-2.0. PanDerm is CC-BY-NC-ND 4.0 and part of the labelled image corpus is CC-BY-NC; adapted PanDerm weights are not shared.

12. Expected outcomes

- **Scientific.** A controlled answer on whether continued dermoscopy pretraining improves original DINOv2 on external institutions; a negative result remains informative.
- **Engineering.** A reproducible Slurm-based workflow, measured 1/4/8-GPU behaviour, an input-pipeline profile and an evidence-based resource recommendation.
- **Transfer.** A locked, auditable DINOv2-family checkpoint selected by the pre-specified validation-only rule for hospital validation, with training and inference environments clearly separated.

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